DISPUTE AGENT

OUR RECOMMENDATION

APPROVE A STAFF-ASSIST PILOT

S$400K	≈ 5 mo	90%	100%
value a year (base case)	payback on the build	decisions right on 50 test cases	eligibility calls right

EXECUTIVE SUMMARY

THE PROBLEM	‘Buy’ only protects buyers who trust how disputes end; 2025’s Carousell scam victims paid outside it. Disputes over used goods are judgement calls.
EXISTING OPTIONS	Shopee’s engine and generic refund tools are built for new goods from businesses; nothing fits used goods between individuals.
OUR SOLUTION	Rules in code, AI for judgement, Carousell’s own policy clauses cited for every decision, and a person approving each one.
FEASIBILITY (CTO)	Working prototype: about 9 s and under S$0.01 of AI cost per case, 69 automated tests, standard Python stack.
DATA AND EVIDENCE	On 50 hand-labelled test cases: 90% decisions right (single prompt 88%), 100% eligibility (68%), no invented clauses.
BUSINESS CASE (CFO)	S$400K a year base case (S$71K–S$2.0M); breaks even if 6 in 10,000 more deals use ‘Buy’.
OUR ASK	**Approve a staged staff-assist pilot, measured on time to resolve, decisions overturned and ‘Buy’ usage.**

TRUST IS THE PRODUCT

Carousell's 'Buy' button holds the payment until the item arrives. Buyers stay on it only if they trust how disputes end. And 'not as described' disputes over used goods are judgement calls: a missing charger is covered, "light pink vs rose pink" is not. Get it wrong either way and Carousell loses.

29% of Singapore's 6,703 e-commerce scam cases in 2025 were on Carousell. Victims paid by PayNow or bank transfer, outside 'Buy'

1,555 Carousell scam cases in H1 2026, up 17.1%: the only designated online service to rise

78,000+ user reports Carousell's trust team reviewed in Singapore in 2025

Sources: SPF Annual Scams and Cybercrime Brief 2025; SPF Mid-Year Brief 2026; Carousell Regional Scam Trends Report 2025; Carousell SG help centre (copied 28 Oct 2026).

MANUAL RESOLUTION BECOMES SLOW & INEFFICIENT

When buyer and seller can’t agree, Carousell’s support team reads the listing, both sides’ photos and the chat, and decides. Whoever misses an evidence deadline loses, and the payment stays held throughout.

HOW A DISPUTE WORKS TODAY

(1) BUYER RAISES A DISPUTE before the window closes (e.g. 2 days after tracked delivery)

(2) SELLER HAS 24 HOURS to accept or counter-offer; no reply = buyer’s proposal applies

(3) BUYER HAS 24 HOURS to accept the counter-offer

(4) SUPPORT TEAM DECIDES emails both sides, collects evidence, decides

This is where an AI agent can step in

Source: Carousell SG help centre: Overview of the dispute handling process; Buyer Protection and Certified Refund Policy (copied 28 Oct 2026). Report sections 4.2–4.4.

S$400K A YEAR

BASE CASE, SINGAPORE

S$80K staff time saved
+ S$320K more ‘Buy’ fees
Range S$71K to S$2.0M

HOW WE GET THERE

STAFF TIME
6,000 disputes reach staff a year; 2,250 hours saved at S$35 an hour

TRUST
If fairer disputes move 1 point more deals onto ‘Buy’: 80,000 deals × S$100 × 4% fee

SENSE CHECK
Base case implies S$9.6M of ‘Buy’ fees: 5–7% of Carousell Group revenue

COST AND PAYBACK (ASSUMPTIONS)

Build and integration (amortised over 3 yrs)	**S$150K**
Upkeep a year	**S$50K**
AI usage a year	**≈ S$50**
Net value a year	**≈ S$300K**
Payback	**≈ 5 months**

BREAK-EVEN: +0.06 POINTS

About 6 more ‘Buy’ deals in every 10,000 cover the yearly cost: one-sixteenth of our base assumption.

OTHER OPTIONS DO NOT FIT CAROUSELL'S SITUATION

Carousell's disputes are between individuals over used goods with subjective descriptions. A bank or a new-goods shop doesn't have this problem, so no off-the-shelf tool is built for it.

EXISTING OPTIONS IN THE MARKET, AND WHY THEY FALL SHORT

TODAY'S MANUAL REVIEW	Staff read each escalated dispute by hand	**Slow; outcomes may vary between staff**
SHOPEE'S AI RETURN ENGINE	Decides return disputes faster	**Built for new goods from businesses; not for sale**
GENERIC AI REFUND TOOLS	Apply a shop's own return policy	**Made for shops' own stock, not used goods between two people**
TAOBAO PUBLIC JURY	Volunteer users vote on disputes	**Needs a huge user base; seller jurors favour sellers**
OUR DISPUTE AGENT	Carousell's own policy, cited for every decision	**Built for used goods; a person approves every case**

Sources: Sea Limited Q4 2024 earnings call (Shopee); vendor sites (e.g. Fini); report section 4.6. Carousell already applies AI to support workflows, but not to disputes.

HOW IT DECIDES

Rules in code where rules are exact, AI for judgement, and a person for every final call.

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
INTAKE	**ELIGIBILITY**	**CLAIM MATCHING**	**POLICY SEARCH**	**JUDGEMENT**	**CHECK**	**STAFF DECIDE**	**MESSAGES**
Loads listing, order, complaint, reply and chat	Paid with ‘Buy’? In the window? Each check cites its clause	Pairs what the listing promised with what the buyer reports	One search per gap, up to 8 clauses	Outcome, refund %, confidence, clause IDs	Drops unretrieved citations; escalates if unsure	Approve or override; every decision logged	Drafts to buyer and seller
BY CODE	**BY CODE**	**BY AI**	**BY RAG**	**BY AI**	**BY CODE**	**BY PERSON**	**BY AI**

100% eligibility right (single AI prompt: 68%)

6% escalated to a person, including 2 of its 5 mistakes

0 invented clause IDs in final answers, with every model

Source: report sections 5 and 8.2 (Gemini judge, 50 held-out test cases, hand labels). Workflow: src/dispute_agent/agent/graph.py (LangGraph, with an interrupt for the staff decision).

FOUR DESIGN CHOICES

Each choice answers a question you would ask before trusting it with real money.

(1)

RULES IN CODE

Payment method, dates and tracking are exact, so plain code decides them and costs nothing.

100% eligibility vs 68% for a single AI prompt

(2)

EVERY DECISION CITES A CLAUSE

The judge must cite policy clause IDs; code rejects any clause it didn't retrieve, then escalates.

0 invented clause IDs in final answers, with every model

(3)

THE JUDGE CAN SEARCH AGAIN

If the clauses don't cover a gap, it asks for a search (search_policy); code runs it and loops back once.

A safety net: never needed on the test set

(4)

A PERSON DECIDES

Both sides are individuals with money at stake, so staff approve or override every recommendation.

Our partial-vs-return rubric fills a gap in Carousell's policy

Source: report section 5.3 and 8.2. The tool call is structured JSON, so it works the same with Gemini, Ollama and OpenAI. Rubric: DECISION_GUIDE in agent/prompts.py.

UNSURE? A PERSON DECIDES

When the agent isn’t sure, it doesn’t guess. It sends the case to staff with follow-up questions, such as asking the buyer for photos of the outer packaging and shipping label.

A CASE GOES TO A PERSON IF...

(1) Confidence is below 0.7

(2) It cites a clause it wasn’t given, or no valid citation is left

(3) It says more evidence is needed

(4) A gap between listing and complaint is marked unclear

(5) The outcome contradicts its own gap assessment

(6) It still wants policy the search could not find

6%
of test cases escalated

2 of 5
of its mistakes were among them

94%
right on the cases it does not escalate

GROUNDED IN CAROUSELL'S OWN POLICY

Four Carousell Buyer Protection pages, split into 37 clauses with stable IDs like BPR-03. Every recommendation must cite them, and code rejects any citation it didn't retrieve, so each decision rests on a rule staff can check, and a policy change is just re-indexing the text.

RIGHT CLAUSE RANKED FIRST (25 TEST QUESTIONS)



96% hybrid search has the right clause in its top five

88% of citations (60 of 68) support the decision they back up (checked by a person)

Sources: reports/retrieval.md (hybrid = ChromaDB embeddings + BM25 keywords, rank fusion); report sections 6.3, 8.4 and 8.6.

HOW THE SEARCH WORKS

Retrieval-augmented generation (RAG) over Carousell's own help pages: a search per gap, not one search per case.

(1) CORPUS

4 Carousell SG help pages, about 24,000 characters, copied by hand on 28 Oct

(2) CHUNK

37 clauses of 300–900 characters, each with a stable ID such as BPR-03

(3) INDEX

ChromaDB with local MiniLM embeddings (meaning) plus BM25 (keywords)

(4) QUERY

One search per claim gap, plus one for the dispute reason

(5) FUSE

Reciprocal rank fusion; top 3 per query, up to 8 clauses per case

76% right clause ranked first (either method alone: 52%)

88% right clause in the top 3 the agent uses

96% right clause in the top 5

0.81 mean reciprocal rank (meaning 0.67, keywords 0.71)

Source: report section 6.3; reports/retrieval.md (25 test queries). The index rebuilds itself when the policy text changes.

WHAT RETRIEVAL REALLY ADDS

We tested whether retrieval changes the decisions or only the citations. On our test set the judge mostly knows the rules already. Retrieval's value is grounding: every decision rests on a clause staff can check, coverage the model would miss is put in front of it, and a policy change is handled by re-indexing the text.

DECISIONS RIGHT, BY RETRIEVAL DESIGN

DESIGN	LLAMA 3.1 8B	QWEN 3 8B	WHAT HAPPENS
Per gap (default)	**54%**	**52%**	One search per gap before judging
Agent-driven	**54%**	**52%**	A planning step calls the search; one more AI call
No retrieval	**64%**	**56%**	Nothing real to cite: every judged case escalated

GEMINI, FIRST 25 TEST CASES

23 of 25 right with or without the policy restated in the prompt

Without retrieval it declined two missing-accessory claims the policy covers (G008, G043), and could cite nothing, so every case went to a person.

Source: report section 8.6 (retrieval and decision-guide ablations; local models are free to run, so Gemini was probed on 25 cases to save quota). A larger policy corpus would make retrieval matter more.

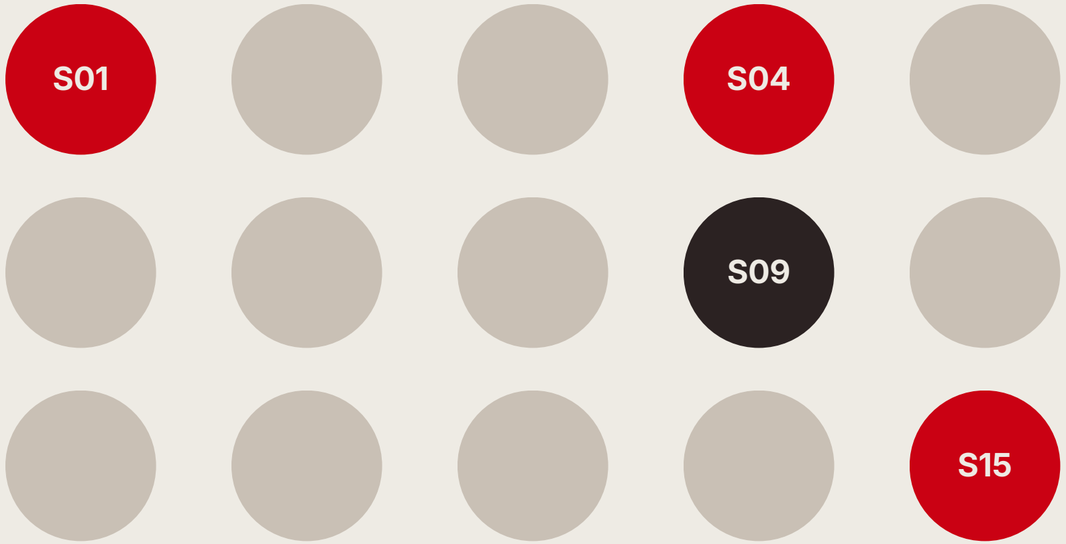
TEST A RULE BEFORE IT SHIPS

PROPOSED RULE

“If a promised accessory is missing, the buyer gets a full refund with return instead of a partial refund.”

The agent re-judges past disputes with the rule added. Past results are cached, so each rule costs one AI call per case.

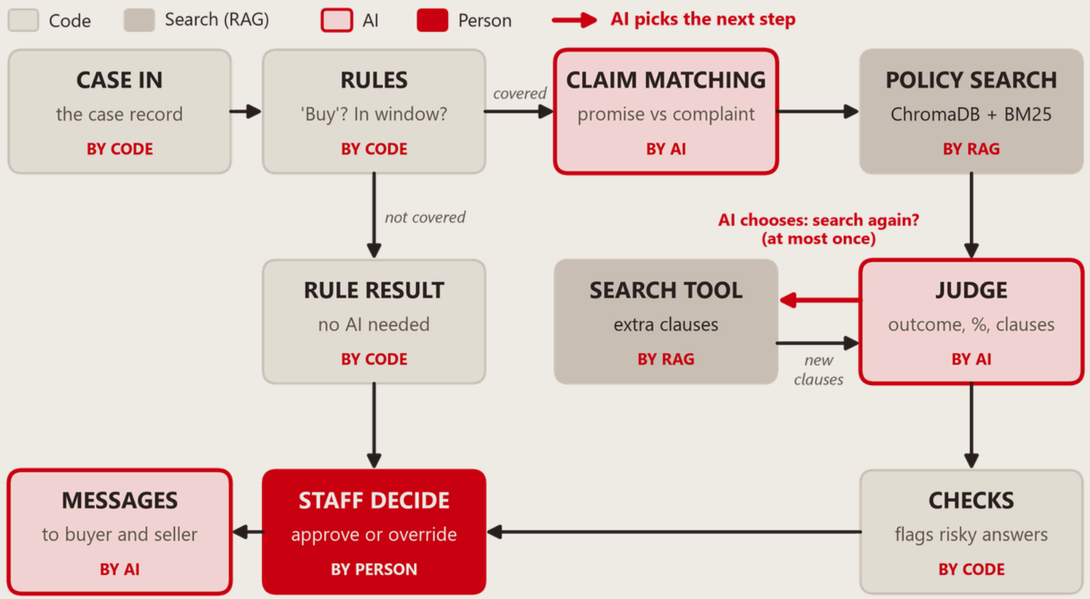
4 OF 15 past decisions would change



- As intended: a promised accessory was missing
- Wrong: that listing never promised it, so a person reviews every change
- No change

Source: what-if test on 15 seed cases (report sections 5.2 and 8.4; src/dispute_agent/evaluation/whatif.py). AI answers can vary between runs.

ARCHITECTURE AND MODELS



Built with LangGraph · ChromaDB + BM25 · Pydantic schemas · Streamlit

WHICH MODEL DOES WHAT

JUDGE (DEFAULT)	**Gemini Flash-Lite, Google cloud** 90% right; 9 s a case; free tier today, paid tier for the pilot
JUDGE (LOCAL OPTION)	**Llama 3.1 8B via Ollama** 54% right: data stays on-site, but not good enough yet
FALLBACK	**OpenAI GPT-5.4 mini** Only if a Gemini call fails; not tested on our cases, so its answers are labelled ‘fallback’
UPGRADES	**Gemini version pinned for the pilot** The 50 test cases are re-run before any model change

BUILT. TESTED. CHEAP.

WHAT IT TAKES TO RUN

SPEED	Measured on the free tier, calls 4.5 s apart
VOLUME	About 25 disputes reach staff a working day
LOCK-IN	Portable code; only Gemini is accurate enough yet
DATA TODAY	Public listings and made-up disputes only

9 s

median time to a cited recommendation

1.8

AI calls per case; ineligible cases need none

< S$0.01

AI cost per case: about S$50 a year in the base case

69

automated tests, run offline without an API key

PATH TO PRODUCTION Integration and privacy work, not research. The next slide shows what the S$150K build adds.

HOW IT PLUGS IN

One case in, one checked recommendation out. PILOT BUILD steps are the S$150K build.

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CASE IN	**PRIVACY**	**WORKFLOW**	**POLICY INDEX**	**AI MODELS**	**RESULT**	**STAFF APP**	**CASE OUT**
Sends one JSON case record: listing, order, complaint, chat	Names, contact and bank details removed before any AI call	LangGraph: rules, claim matching, search, judge, checks	37 clauses; ChromaDB and BM25 rankings fused	Gemini; OpenAI fallback; Ollama on-site. Every answer schema-checked	JSON: outcome, refund %, confidence, clause IDs, escalate flag	Approve or override; the pilot build adds staff login and an audit log	Decision and message drafts go back to your case system
PILOT BUILD	**PILOT BUILD**	**RUNS TODAY**	**RUNS TODAY**	**RUNS TODAY**	**RUNS TODAY**	**DEMO TODAY**	**PILOT BUILD**

6.6 s S01, a real run, end to end: 1 gap found, 2 AI calls

15% S01: 15% refund (S$27.54) at confidence 0.9, citing BPR-02, DPO-08, DPO-02

6 mo 2 engineers, S$150K: build the PILOT BUILD steps, then run the 14-week pilot

Source: S01 from outputs/runs/gemini_seed/agent.jsonl (Gemini Flash-Lite). Build cost: report section 9.6 (an assumption). Code: src/dispute_agent/agent/graph.py, llm/, rag/.

BUILT TO FAIL SAFELY

A support tool has to keep working when an API is slow, out of quota or down. Every failure path ends with a person or a clear message, never a silent wrong answer.

DEPLOYMENT STATUS
Public demo on Streamlit Community Cloud since 6 Nov, made-up data only. Keys in Streamlit secrets; Gemini first, OpenAI as fallback.

WHAT HAPPENS WHEN THINGS GO WRONG

RATE LIMITS	Gemini calls are paced at least 4.5 s apart; short limits are retried
QUOTA RUNS OUT	Long jobs stop cleanly and resume where they stopped
GEMINI FAILS	OpenAI takes over; untested on our cases, so its answers are labelled ‘fallback’ for close checking
DATA MUST STAY ON-SITE	A local model runs via Ollama, but at 54% right it isn’t ready for the pilot
BOTH PROVIDERS DOWN	No recommendation is made; staff handle the case as they do today
CODE CHANGES	69 automated tests run offline, with fake AI responses

LIVE DEMO

TRY IT YOURSELF
https://bc3415-carousell-dispute-agent.streamlit.app/

(1)

S01 · PS VITA SLIM

A PROMISED PART IS MISSING

WATCH FOR Partial refund of 15%, citing the clauses on incomplete items. Staff approve; messages drafted.

(2)

S16 · ANIME DVD

PAID BY BANK TRANSFER

WATCH FOR Rules say ‘not covered’ straight away, with no AI cost.

(3)

G155 · NINTENDO GAMES

THE EVIDENCE CONFLICTS

WATCH FOR The agent isn’t sure, so it goes to a person with a follow-up question for the buyer.

SWITCHING TO THE LIVE APP

200 DISPUTES, BUILT TO TEST

REAL LISTINGS	300 real second-hand listings from Mercari; prices in S$
FACTS SET BY CODE	Payment method and dates come from code, so eligibility labels are exact
PEOPLE'S WORDS BY AI	Complaint, reply and chat (even Singlish), quoting the listing text
LABELLED BY HAND	50 test cases labelled by our team; an AI reviewer agreed on 47 of 50

17 SCENARIOS IN A 60 / 30 / 10 MIX

CLEAR · 120	BORDERLINE · 60	INELIGIBLE · 20
Missing accessory 18	Minor shade difference 12	Paid outside 'Buy' 5
Wrong size, colour or model 18	"Feels like 6/10" 12	Raised too late 5
Undisclosed defect 18	Wear the listing disclosed 13	Order already completed 5
Condition misrepresented 16	Small flaw on a 'like new' item 12	Untracked mail not received 5
Change of mind 17	Conflicting accounts 11	
Not in the listing 17		
Damage after delivery 15		
Seller didn't respond 1		

TEST SET 50 cases (30 clear, 15 borderline, 5 ineligible) that nobody tunes prompts on; the other 150 are for development.

Source: Mercari Price Suggestion dataset, Kaggle; seed 3415. Labels: B1 and B2 labelled 25 test cases each on 4 Nov; AI reviewer kappa 0.90.

A TIE ON JUDGEMENT. A WIN ON THE RULES.

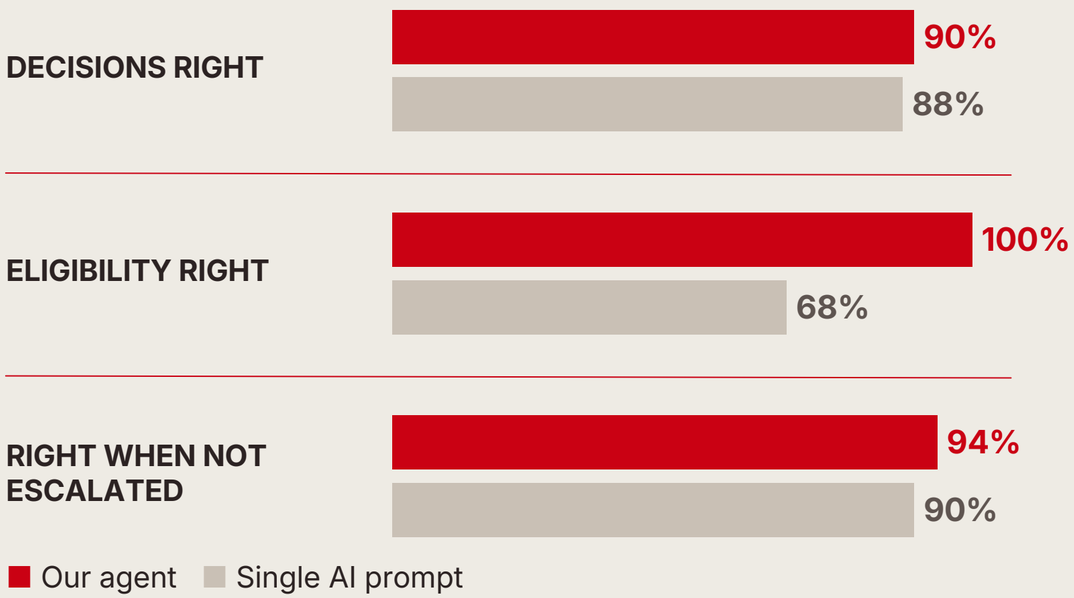
THE DATA

300 real second-hand listings (Mercari)

200 disputes written on top: 20 by hand, 180 by a different model from the judge

50 held-out test cases, labelled by hand by the team

AGENT VS A SINGLE AI PROMPT WITH THE WHOLE POLICY (GEMINI)



READ IT HONESTLY

One case is 2 points, so 90% vs 88% is a tie. The gain is on the rules, the errors that cause appeals.

It leans to declining: 96% of seller wins right, 83% of buyer wins. We track that.

Source: report section 8.2 (run cmp_gemini-flash-lite, 50 held-out cases, hand labels; exact McNemar p = 1.00). Listings: Mercari Price Suggestion dataset (Kaggle).

WHERE IT WENT WRONG

CASE	CORRECT ANSWER	AGENT SAID	WHAT HAPPENED
G008	Partial refund	Full refund with return	Not escalated; the missing knobs could be judged either way
G059	Full refund with return	Decline	Escalated to a person; a judgement call (the AI reviewer also said decline)
G125	Decline	Partial refund	Escalated to a person; the single prompt made the same mistake
G129	Partial refund	Decline	Judgement call: a scratch on the container only
G147	Partial refund	Decline	Judgement call: a ‘6/10’ complaint about a ‘like new’ item

FULL SCORECARD (AGENT VS SINGLE AI PROMPT, GEMINI)

92% vs 88%
refund vs decline direction right

83% vs 96%
buyer-win cases right

96% vs 81%
seller-win cases right

40% vs 17%
of wrong answers escalated

9.1 s vs 4.5 s
median time per case

Source: report sections 8.2 and 8.3 (run cmp_gemini-flash-lite, hand labels). G059, G129, G134 and G147 are cases where reasonable people could disagree; three of them are among the mistakes above.

WHY GEMINI JUDGES

Same 50 test cases, prompts and code. Only the judge model changed.

GEMINI FLASH-LITE

Google cloud, free tier

90%

of decisions right. 9 s a case. Refunded 1 of 21 cases that should be declined.

LLAMA 3.1 8B

Runs on our own PC

54%

of decisions right. 15 s a case. Refunded 19 of 21 that should be declined.

QWEN 3 8B

Runs on our own PC

52%

of decisions right. 19 s a case. Refunded 20 of 21 that should be declined.

THE AGENT WAS AHEAD OF A SINGLE PROMPT WITH EVERY MODEL (+2, +20 and +8 points; none statistically significant). Small local models refund almost everything, so they're only an option where data must stay on-site.

Source: report section 8.5 and reports/model_comparison.md. Local models ran at 4-bit on an 8 GB GPU; larger open models might close the gap.

WHAT COULD GO WRONG

Each risk is measured on our test set and handled by design, not by hope.

RISK	HOW WE HANDLE IT	EVIDENCE SO FAR
INVENTED POLICY	Must cite retrieved clauses; code drops others and escalates	**0% in final answers, all 3 models**
BUYER VS SELLER BIAS	Accuracy reported separately for each side	**Leans to declining: 83% vs 96%**
STAFF RUBBER-STAMPING	Confidence shown; unsure cases escalated; overrides logged	**6% escalated, incl. 2 of 5 mistakes**
SYNTHETIC TEST DATA	Different models write and judge; hand labels	**Pilot tests on real cases first**
PRIVACY	Remove personal details; paid tier or local model	**Prototype uses public or made-up data only**

WHAT WE CAN'T PROVE YET

We'd rather you hear our limits from us. Each one is something the pilot measures on Carousell's real disputes.

SYNTHETIC DISPUTES	The generator knew the intended outcome, so cases may be easier than real ones
SMALL TEST SET	50 cases: one case moves accuracy 2 points; no gap is statistically significant
ONE LABELLER PER CASE	No agreement score between labellers yet
TEST SET REUSED	Used three times; nothing was tuned on it, but numbers are slightly optimistic
ANSWERS VARY	Even at temperature 0: G155 changed outcome between runs (escalated every time)
NOT CAROUSELL DATA	Listings are from Mercari (US); local models were limited to 8B by an 8 GB GPU

HOW WE KEPT THE TEST FAIR

Our first test run lost to the single prompt. Instead of tuning until the test set looked good, we fixed the cause on separate development cases, then re-ran the test, and we report every run.

THE RULE WE FOLLOWED

Fixes are chosen on development cases only, never on the test set.

EVERY TEST RUN, SCORED AGAINST THE HAND LABELS

RUN 1	**82% vs 84%**	Agent lost: both declined every small flaw on ‘like new’ items, because two lines of our guide contradicted each other
THE FIX	**dev only**	Clarified the rule and checked it on 36 borderline development cases: agent 81% → 89%
RUN 2	**88% vs 90%**	Re-ran the test set once: the fix helped both systems
RUN 3	**90% vs 88%**	With the search tool added; the headline result in this deck

WHAT'S NEXT: APPROVE THE PILOT

SCAN TO TRY THE LIVE APP

A staff-assist pilot with Gemini as the judge: not automatic decisions, and not a small local model.

A STAGED PILOT (OUR PROPOSAL)

(1)	VALIDATE	A person checks the citations; double-label cases; a timing study replaces the staff-time assumptions	**2 weeks**
(2)	SHADOW MODE	The agent recommends silently beside staff on real cases, personal details removed	**4 weeks**
(3)	STAFF-ASSIST	Staff see recommendations and approve or override every case	**8 weeks**
(4)	OTHER MARKETS	Buyer Protection also runs in HK, MY, TW and VN	**After review**

DECISION REQUESTED: APPROVE THE PILOT

Measured on time to resolve a dispute, decisions overturned on appeal, and share of orders paid with 'Buy'